

ENVS 193DS Final

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Link to Github repo

https://github.com/hparsons333/ENVS-193DS_spring-2026_final

Setting up

```
library(tidyverse)
library(here)
library(MuMIn)
library(ggeffects)
library(DHARMA)
library(janitor)
library(readxl)
```

Problem 1. Research writing

You're working on a research team trying to understand how restoration of native plant cover (measured in percent cover) influences probability of American Avocet (*Recurvirostra americana*) and other shorebird habitat use in the San Francisco Bay. Your co-worker runs some analyses and writes up a report, giving it to you to review. In part 1 of the results section of the report, your co-worker has written:

We rejected the null hypothesis that distance to water edge does not predict probability of American Avocet microhabitat use ($p = 0.03$).

In part 2 of the results section of the report, your co-worker has written:

We rejected the null hypothesis that there is no relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat) ($p < 0.001$).

It's great that this report is coming together, but you think they can improve on what they've written and make it more understandable to a non-statistical audience.

1a. Transparent statistical methods

What statistical tests did your co-worker use?

Clearly connect the test to the part that you are addressing (e.g. "In part 1, they used _____. In part 2, they used _____").

In part 1, they used a simple linear regression to test if distance to water edge predicted American Avocet microhabitat use. In part 2, they used a correlation coefficient test to see if there was a relationship between bird species and habitat.

1b. Suggestions for rewriting

In 1-3 sentences, write new statements with interpretation only for the statements in part a and b.

Be sure your interpretation makes sense without any statistical summary (which, again, you should not include).

Rewriting statement a

Our results suggest that we can reject the null hypothesis that distance to water edge does not predict probability of American Avocet microhabitat use. Alternatively, we determine that the distance to water edge significantly predicts whether or not American Avocet will utilize microhabitats.

Rewriting statement b.

Our results suggest that we can reject the null hypothesis, which states that there is no relationship between the type of bird species found and the habitat type. Alternatively, we determine that there is a significant relationship between the type of bird species found and the habitat type.

1c. Further analysis

In 1-2 sentences, outline:

- one additional variable that could influence the probability of American Avocet microhabitat use
- what type of variable it is (with units and/or levels, if applicable)
- why that variable might influence American Avocet habitat use.

Be specific with units, data collection method, etc. as needed.

Because American Avocet need their nesting, brooding-rearing, foraging, and roosting sites to be protected from predators, another variable that could influence the probability of American Avocet habitat use is island distance from landmass/shore, since increased distance from the nearest landmass would provide increased distance from larger predators.

This would be a continuous variable, measured in distance in kilometers (since the area of the pond is measured in kilometers squared).

Problem 2. Data analysis

Reading in data

```
# creating new object
nests <-
  # reading in csv
  read_csv(
    # directing function to data folder
    here("data",
        # csv: nest_data_final
        "nest_data_final.csv"))
```

2a. Clean and wrangle

```
# creating new object
nests_clean <-
  # df: nests
  nests |>
  # selecting only necessary columns: tree height, bird nest occurrence, and ant species pres
```

```

select(height_cm, case_control, ant_species) |>
# filtering ant_species column to not include NA or T. penzigi
filter(ant_species == "RRB" | ant_species == "AB" | ant_species == "BBR") |>
# creating new column to store ant species' binomial names
mutate(ant_species_name = case_match(
  ant_species,
  "RRB" ~ "Crematogaster mimosae", # C. mimosae
  "AB" ~ "Crematogaster sjostedti", # C. sjostedti
  "BBR" ~ "Crematogaster nigriceps")) |> # C. nigriceps
# reordering ant species levels
mutate(ant_species_name = as_factor(ant_species_name),
       ant_species_name = fct_relevel(ant_species_name,
                                       "Crematogaster mimosae",
                                       "Crematogaster sjostedti",
                                       "Crematogaster nigriceps"))

```

Warning: There was 1 warning in `mutate()`.

i In argument: `ant\_species\_name = case\_match(...)`.

Caused by warning:

! `case\_match()` was deprecated in dplyr 1.2.0.

i Please use `recode\_values()` instead.

```

# creating new object
nests_clean_display <-
  # df: nests_clean
  nests_clean |>
  # selecting 10 random observations
  slice_sample(n = 10)

# displaying object
nests_clean_display

```

```

# A tibble: 10 x 4
  height_cm case_control ant_species ant_species_name
    <dbl>      <dbl> <chr>      <fct>
1     201          1 BBR      Crematogaster nigriceps
2     243          0 RRB      Crematogaster mimosae
3     185          0 RRB      Crematogaster mimosae
4     187          0 RRB      Crematogaster mimosae
5     276.          0 RRB      Crematogaster mimosae
6     187          0 RRB      Crematogaster mimosae

```

7	472	1 RRB	Crematogaster mimosae
8	512.	1 RRB	Crematogaster mimosae
9	246	1 BBR	Crematogaster nigriceps
10	159	0 RRB	Crematogaster mimosae

```
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : chr [1:270] "RRB" "RRB" "RRB" "RRB" ...
 $ ant_species_name: Factor w/ 3 levels "Crematogaster mimosae",...: 1 1 1 1 2 1 3 1 2 3 ...
```

2b. Response variable

In 1-2 sentences, explain what the 1s and 0s mean in this data set biologically.

A 1 value in the case_control column corresponds to a tree that was observed to have a birds nest. A 0 value corresponds to a tree that did not have a birds nest, or was “available” for one.

2c. Purpose of study

In 2-3 sentences for each predictor, describe:

- what kind of variable it is (i.e. continuous, discrete, categorical, etc.) - how it could possibly influence the probability of nest occurrence (i.e. as tree height increases, the probability of probability of nest occurrence would _____ because...) - where in the paper you found this information (the specific section, subsection, and paragraph)

The type of ant species found is a categorical variable. Nest occurrence is predicted to be influenced by the aggression of the ant species found in each tree, and nest occurrence will increase with more aggressive species of ants found, because ants will indirectly protect the nests from predation. I found information about the hypothesis in fourth paragraph of the introduction, the collection of the data for ants in the first paragraph of the methods. and the results for the model of ant species as a predictor in the third paragraph of the results.

Tree height is continuous data. Nest occurrence may be influenced by tree height in that as tree height increases, nest occurrence probability will increase, as being further from the ground provides more protection from predators. I found the information about how tree height data was collected in the first paragraph of the methods, the results for the model of tree height as

a predictor in the second paragraph of the results section, and information on what type of data it was in the table of the results section.

2d. Table of models

Model number	Ant species	Tree height	Predictor list
0	X	X	all predictors (full model)
1		x	tree height W/interaction effect of ant species

2e. Run the models

```
# model 0: all predictors
model0 <- glm(
  case_control ~ ant_species_name + height_cm, # formula: all predictors
  data = nests_clean, # clean data frame
  family = "binomial") # family is binary response variable

# model 1: tree height as a predictor with interaction effect of ant_species
model1 <- glm(
  case_control ~ height_cm * ant_species_name, # formula: height_cm w/interaction effect of ant_species
  data = nests_clean, # clean data frame
  family = "binomial") # family is binary response variable
```

2f. Select the best model

```
# calculating lowest AIC value between models
AICc(model0,
      model1)
```

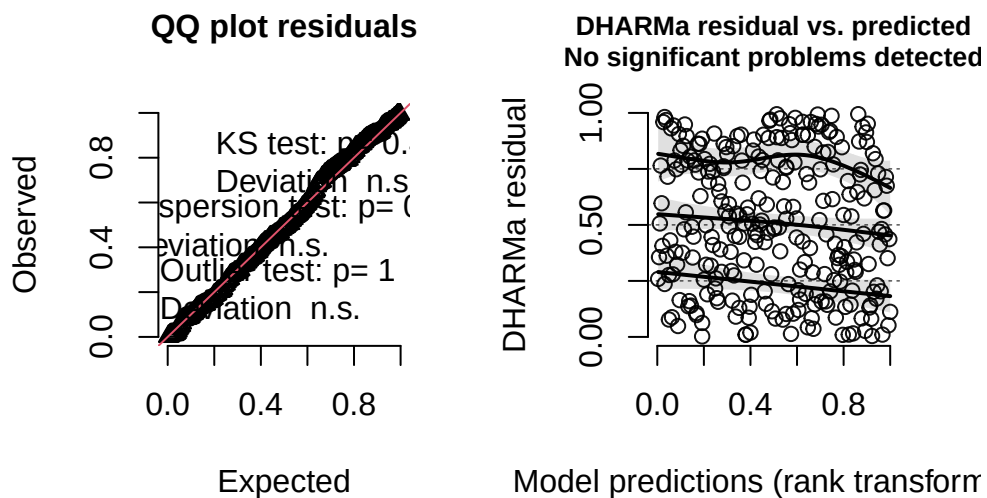
```
      df      AICc
model0  4 258.8940
model1  6 238.1236
```

The best model as determined by Akaike's Information Criterion (AIC) was the model where acacia ant species identity influenced the effect of tree height on probability of nest occurrence.

2g. Check diagnostics

```
# displaying plots in 2x2 grid
par(mfrow = c(2,2))
# generating diagnostic plots
plot(simulateResiduals(model1))
```

DHARMA residual



Note: - The residuals appear to follow a normal distribution - The residuals don't appear to be dispersed from the normal distribution - There doesn't appear to be any outliers

2h. Visualize the model predictions

```
# creating a new object
preds <-
# generating model based estimate
ggpredict(
  # model: model1
  model1,
  # terms: case_control as predicted by effect of height_cm by ant species
  terms = c("height_cm", "ant_species_name")) |>
```

```
# renaming
rename(height_cm = x,
        ant_species_name = group)
```

Data were 'prettified'. Consider using `terms="height\_cm [all]"` to get smooth plots.

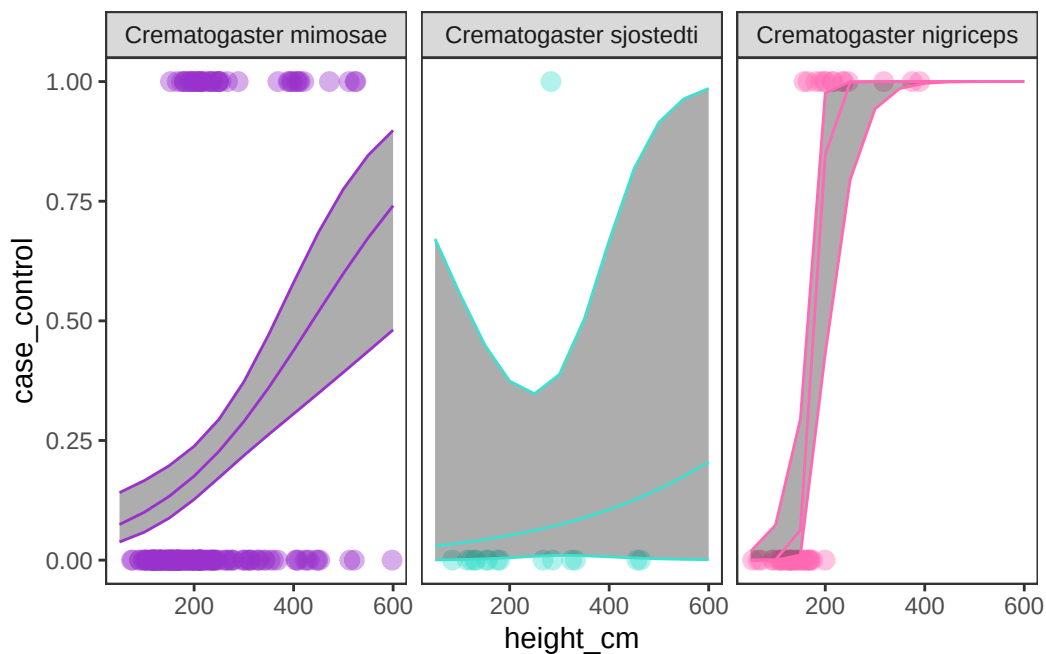
```
# base layer: ggplot
ggplot(
  # df: clean nests data
  data = nests_clean,
  # x-axis: tree height
  # y-axis: bird nest occurrence
  mapping = aes(x = height_cm,
                 y = case_control,
                 # color by ant species
                 color = ant_species_name)) +
# first layer: scatterplot
geom_point(size = 3,
           alpha = 0.4) +
# second layer: ribbon to represent confidence interval
geom_ribbon(
  # df: preds
  data = preds,
  # x-axis: height_cm
  # y-axis: predicted bird nest occurrence
  # y-min: lower bound of confidence interval
  # y-max: upper bound of confidence interval
  aes(x = height_cm,
       y = predicted,
       ymin = conf.low,
       ymax = conf.high),
  # changing ribbon to be transparent
  alpha = 0.4) +
# second layer: model estimate
geom_line(
  # df: preds
  data = preds,
  # x-axis: height_cm
  aes(x = height_cm,
       # y-axis: predicted nest occurrence
```



```

      y = predicted)) +
# changing the theme
theme_bw() +
# separating plots by ant species present
facet_wrap(~ ant_species_name) +
# getting rid of legend
theme(legend.position = "none") +
# changing colors
scale_color_manual(values = c("darkorchid", "turquoise", "hotpink")) +
# getting rid of grid lines
theme(panel.grid.major = element_blank(),
      panel.grid.minor = element_blank())

```



2i. Write a caption for your figure

Figure 1. Effect of tree height on nest occurrence differs by acacia ant species. Small transparent dots represent measurements of nest occurrence at different tree heights in the presence of three different species of ants: *C. mimosae*, *C. sjostedti*, and *C. nigriceps*. Lines represent estimated nest occurrence at different tree heights in the presence of the three ant species. Ribbon represents the confidence interval for the estimate of the model. Data originally from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati

Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

2j. Calculate model predictions

```
summary(model1)
```

Call:

```
glm(formula = case_control ~ height_cm * ant_species_name, family = "binomial",
     data = nests_clean)
```

Coefficients:

	Estimate	Std. Error	z value
(Intercept)	-2.845076	0.434500	-
6.548			
height_cm	0.006492	0.001572	4.130
ant_species_nameCrematogaster sjostedti	-0.848334	2.554617	-
0.332			
ant_species_nameCrematogaster nigriceps	-13.128072	5.405092	-
2.429			
height_cm:ant_species_nameCrematogaster sjostedti	-0.002597	0.008386	-
0.310			
height_cm:ant_species_nameCrematogaster nigriceps	0.081944	0.030921	2.650

Pr(>|z|)

(Intercept)	5.83e-11 ***
height_cm	3.63e-05 ***
ant_species_nameCrematogaster sjostedti	0.73983
ant_species_nameCrematogaster nigriceps	0.01515 *
height_cm:ant_species_nameCrematogaster sjostedti	0.75679
height_cm:ant_species_nameCrematogaster nigriceps	0.00805 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 286.04 on 269 degrees of freedom
 Residual deviance: 225.80 on 264 degrees of freedom
 AIC: 237.8

Number of Fisher Scoring iterations: 7

```
# generating model based estimate
ggpredict(
  # model: model1
  model1,
  # terms: case_control as predicted by effect of height_cm by ant species
  terms = c("height_cm [600]", "ant_species_name"))
```

Predicted probabilities of case_control

ant_species_name: *Crematogaster mimosae*

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

ant_species_name: *Crematogaster sjostedti*

height_cm	Predicted	95% CI
600	0.20	0.00, 0.99

ant_species_name: *Crematogaster nigriceps*

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

2k. Interpret your results

Bird nest occurrence tends to significantly increase with increasing tree height in the presence of *C. nigriceps* as the trendline visualizes in Figure 1.

C. nigriceps ants are more aggressive than *C. sjostedti* and *C. mimosae*, and birds may be selecting for these trees that are both taller and host the more aggressive acacia ant species, because this increases their protection from predators and therefore their fitness.

In the presence of *C. nigriceps*, with every 1 cm increase in height the odds of a bird nesting in a tree increases by a factor of 1.1 (95% CI: [0,0], $p = 0.008$, $\alpha = 0.05$). In the presence of *C. nigriceps*, with a tree height of 600 cm, the probability of a bird nesting in that tree is 1 (95% CI: [1,1]).

Problem 3. Affective and exploratory visualizations

3a. Comparing visualizations

Compare and contrast your affective visualization from Homework 3 and the exploratory visualizations you made for Homework 2. In 1-3 sentences each, explain:

- How are the visualizations different from each other in the way you have represented your data?

The visualization that I created in homework 2 were very basic, and didn't clearly display a trend or the relationship I was trying to visualize (although this may have also been because I didn't have enough data points yet). The affective visualization was a lot more abstract, and did require the artist statement to be fully understood, but I feel like it represented the relationship I wanted to display from the figures that I choose to represent the data point and the background I chose as well.

- What similarities do you see between all your visualizations?

I chose to use the scatter plot for both my exploratory visualization and my affective visualization, and between all my visualizations there seems to be the most weight on the left side of the axis rather than the right, most likely because both the stress and outside engagement variables increased as time spent on the job search decreased.

- What patterns (e.g. differences in means/counts/proportions/medians, trends through time, relationships between variables) do you see in each visualization? Are these different between visualizations? If so, why? If not, why not?

I mostly see what could be described as a negative linear relationship, or a negative trend, between both my outside engagements and time spent on the job search, and my level of stress and time spent on the job search. These are similar because when I am stressed or when I have a lot of outside engagements, I either feel less motivated to spend time on a job search (because I am concerned about other work that I have) or because I do not have the time available to apply.

- What kinds of feedback did you get during week 9 in workshop or from the instructors? How did you implement or try those suggestions? If you tried and kept those suggestions, explain how and why; if not, explain why not.

The two suggestions that I got from workshop and implemented were to use briefcases as representations of my data points, and to color in the balloons to represent decreasing and increasing levels of stress. These were both extremely helpful, as using briefcases was more effective at representing the response variable I had chosen, and coloring the balloons by stress allowed me to get rid of visual clutter (where I initially created a color gradient below the data

points) and represent increasing feelings of stress through the transition of cool colors to warm colors.

3b. Designing an analysis

Based on your response variable and one of your potential predictor variables (your focal variable or otherwise), propose one analysis that we talked about this quarter that you could apply to answer your research question(s).

I want to test if there is actually a difference in the time spent on my job search among different levels of stress, so I will run a Kruskal-wallis test.

Additionally, explain why this analysis is appropriate with regard to:

- your question

This analysis is appropriate in regards to my question because I have multiple groups that I am testing (levels of stress), and I don't have normally distributed data.

your variables and what types of variables they are

My variables are my response variable, which is discrete data counted as the time I spent on the job search in minutes, and my explanatory variable, which is categorical data recorded as the level of stress I felt on the day I took my observation.

3c. Check your assumptions and run your analysis

Creating an object for personal data

```
# creating new object
personal_data <-
  # reading csv
  read_csv(
    # directing function to data folder
    here("data",
      # csv: Personal Data - Sheet1.csv)
      "personal-data.csv"))
```

Rows: 41 Columns: 7

-- Column specification -----

Delimiter: ","

chr (1): date

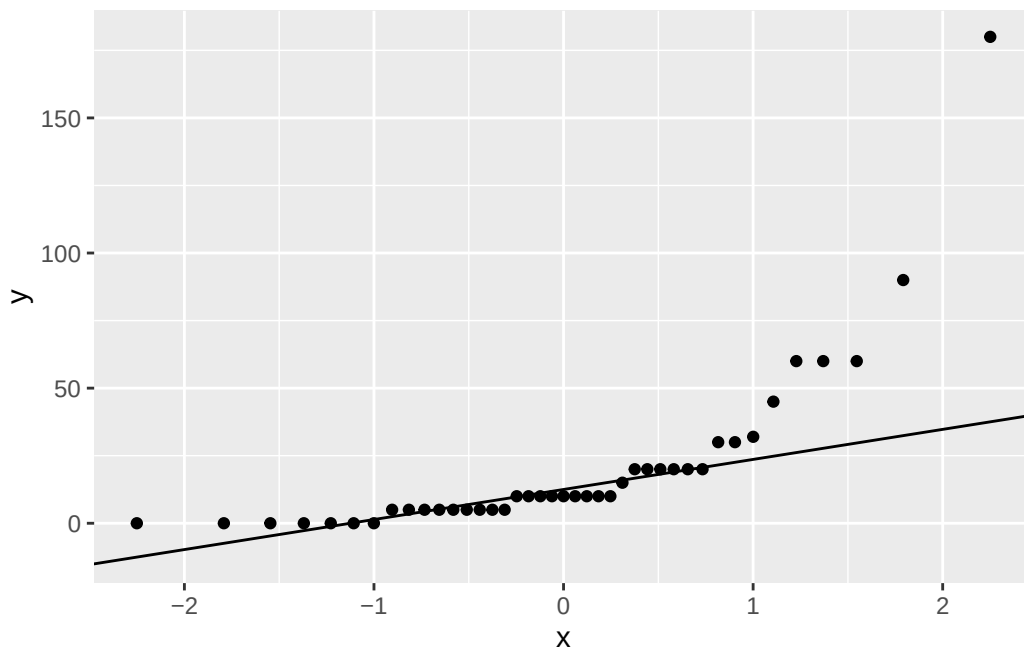
dbl (6): job_search, outside_eng, stress, sleep, social_media, day_to_grad

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

Checking distribution of job_search and stress

```
# creating new object
qqplot_job <-
  # baselayer: ggplot
  ggplot(
    # df: personal_data
    data = personal_data,
    # variable: job_search
    aes(sample = job_search)) +
  # first layer: distribution of temp observations
  stat_qq(distribution=qnorm) +
  # second layer: expected distribution of time spent on job search
  stat_qq_line( col = "black")

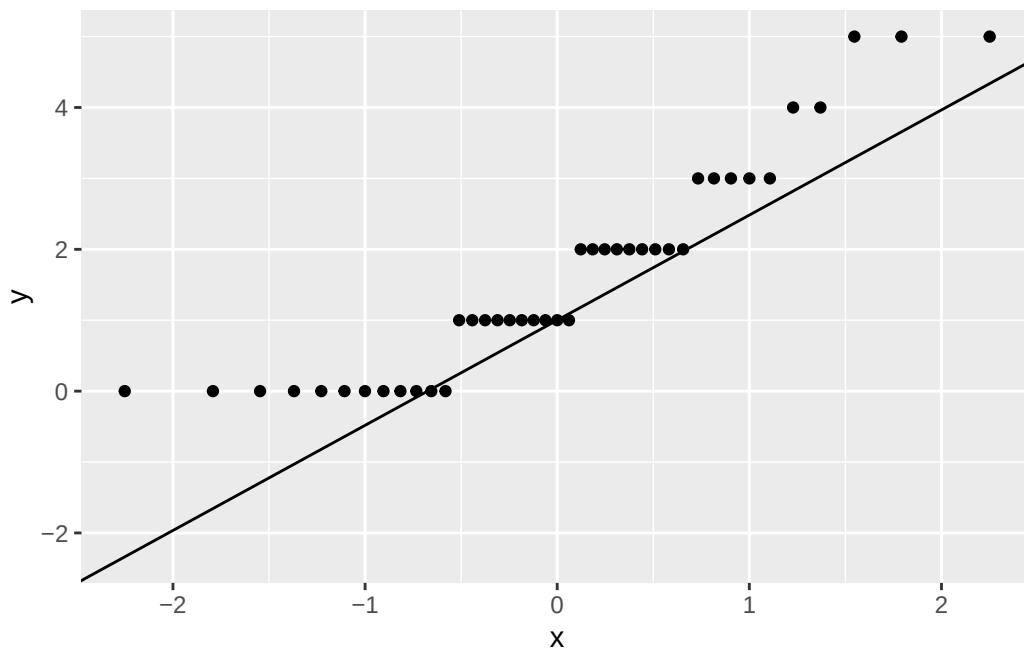
# displaying object
qqplot_job
```



Note: not really normally distributed

```
# creating new object
qqplot_stress <-
  # baselayer: ggplot
  ggplot(
    # df: personal_data
    data = personal_data,
    # variable: job_search
    aes(sample = stress)) +
  # first layer: distribution of temp observations
  stat_qq(distribution=qnorm) +
  # second layer: expected distribution of time spent on job search
  stat_qq_line( col = "black")

# displaying object
qqplot_stress
```

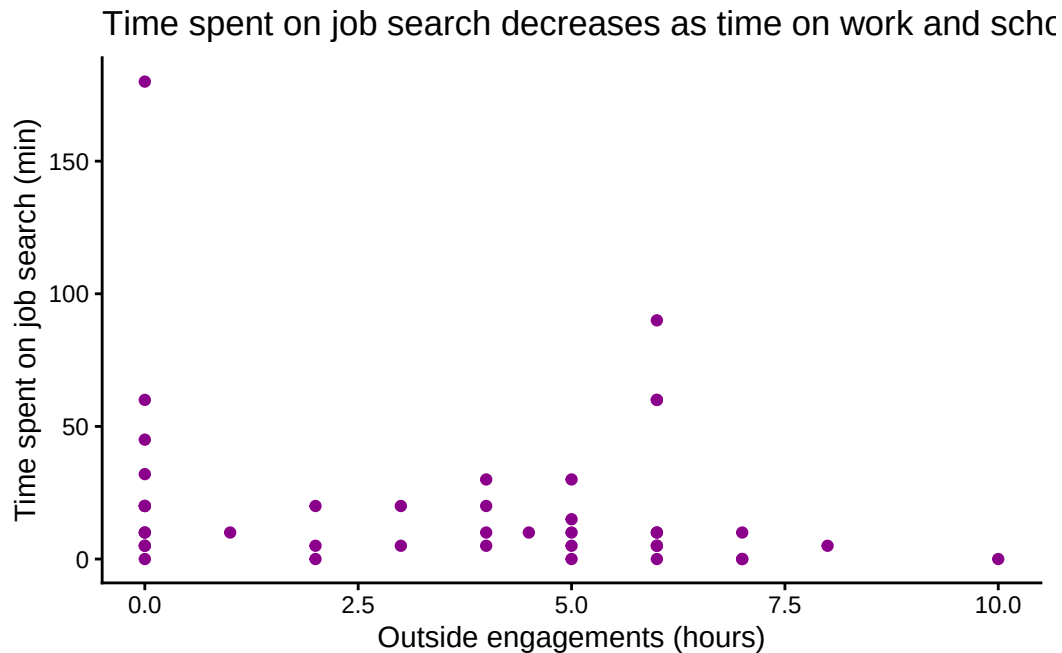


Note: definitely not normally distributed, so Spearman's ρ is required.

Displaying visual relationship

```
# creating a new object
visual_test <-
  # base layer: ggplot
  ggplot(
    # df: personal_data
    data = personal_data,
    # x-axis: outside_eng
    # y-axis: job_search
    mapping = aes(x = outside_eng,
                  y = job_search)) +
  # first layer: scatterplot
  geom_point(
    # changing color of points
    color = "magenta4") +
  # changing theme
  theme_classic() +
  # changing title and axes' titles
  labs(title = "Time spent on job search decreases as time on work and school increases",
       x = "Outside engagements (hours)",
       y = "Time spent on job search (min)")

# displaying object
visual_test
```

Appears mostly monotonic, in that it decreases in one direction

Checklist of assumptions

1. There is a monotonic relationship
2. The observations are independent of one another

Running a predictive model

```
#correlation test
model2 <- lm(
  job_search ~ outside_eng, # formula: all predictors
  data = personal_data) # clean data frame

model2
```

Call:

```
lm(formula = job_search ~ outside_eng, data = personal_data)
```

Coefficients:

```
(Intercept)  outside_eng
      29.159      -2.327
```

```
# displaying plots in 2x2 grid
par(mfrow = c(2,2))
# generating diagnostic plots
plot(model2)
```

